Question 2 - STAT 4C03

```
data = read.csv("Admissions.csv")
head(data)
```

```
##      Admit Gender Dept Freq
## 1 Admitted   Male    A  512
## 2 Rejected   Male    A  313
## 3 Admitted Female    A   89
## 4 Rejected Female    A   19
## 5 Admitted   Male    B  353
## 6 Rejected   Male    B  207
```

b)

```
tab.a <- xtabs(Freq ~ Gender + Admit + Dept, data = data)
tab.a
```

```
## , , Dept = A
##
##         Admit
## Gender   Admitted Rejected
##   Female       89       19
##   Male        512      313
##
## , , Dept = B
##
##         Admit
## Gender   Admitted Rejected
##   Female       17        8
##   Male        353      207
##
## , , Dept = C
##
##         Admit
## Gender   Admitted Rejected
##   Female      202      391
##   Male        120      205
##
## , , Dept = D
##
##         Admit
## Gender   Admitted Rejected
##   Female      131      244
```

```
##   Male        138      279
##
## , , Dept = E
##
##         Admit
## Gender   Admitted Rejected
##   Female       94      299
##   Male         53      138
##
## , , Dept = F
##
##         Admit
## Gender   Admitted Rejected
##   Female       24      317
##   Male         22      351
```

```r
# Conditional independence Gender/Admission
model.a.1 <- glm(Freq ~ Gender + Admit + Dept + Gender:Dept + Admit:Dept,
          family = poisson, data = data)

# Homogeneous association adding Gender:Admit
model.a.2 <- glm(Freq ~ Gender + Admit + Dept + Gender:Admit + Gender:Dept + Admit:Dept,
          family = poisson, data = data)

# Test: is Gender:Admit needed? model.a.1 compared model.a.2
anova(model.a.1, model.a.2, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Freq ~ Gender + Admit + Dept + Gender:Dept + Admit:Dept
## Model 2: Freq ~ Gender + Admit + Dept + Gender:Admit + Gender:Dept + Admit:Dept
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         6     21.735
## 2         5     20.204  1   1.5312   0.2159
```

Conclusion:

Fail to reject H0. Adding Gender:Admit does not significantly improve fit, conditional independence holds. Within each department, gender has no additional effect on admission beyond what is already captured by Gender:Dept and Admit:Dept.

b)

```r
# Separate admitted and rejected
admitted <- data[data$Admit == "Admitted", c("Gender", "Dept", "Freq")]
rejected  <- data[data$Admit == "Rejected", c("Gender", "Dept", "Freq")]

# Merge by Gender and Dept
new_data <- merge(admitted, rejected, by = c("Gender", "Dept"))
names(new_data)[3:4] <- c("Admitted", "Rejected")
head(new_data)
```

```
##   Gender Dept Admitted Rejected
## 1 Female    A       89       19
## 2 Female    B       17        8
## 3 Female    C      202      391
## 4 Female    D      131      244
## 5 Female    E       94      299
## 6 Female    F       24      317
```

```r
# Fit the model
model.b <- glm(cbind(Admitted, Rejected) ~ Gender + Dept,
               family = "binomial",
               data = new_data)
summ.b = summary(model.b)
summ.b
```

```
##
## Call:
## glm(formula = cbind(Admitted, Rejected) ~ Gender + Dept, family = "binomial",
##     data = new_data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.68192    0.09911   6.880 5.97e-12 ***
## GenderMale  -0.09987    0.08085  -1.235    0.217
## DeptB       -0.04340    0.10984  -0.395    0.693
## DeptC       -1.26260    0.10663 -11.841  < 2e-16 ***
## DeptD       -1.29461    0.10582 -12.234  < 2e-16 ***
## DeptE       -1.73931    0.12611 -13.792  < 2e-16 ***
## DeptF       -3.30648    0.16998 -19.452  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 877.056  on 11  degrees of freedom
## Residual deviance:  20.204  on  5  degrees of freedom
## AIC: 103.14
##
## Number of Fisher Scoring iterations: 4
```

The **regression parameters** represent the log-odds ratio relative to the reference group (Female, Dept A).

Summary

The fitted model regresses the probability of admission on Gender and Department using a logistic link. The reference category is Female applicants in Department A.

Intercept (0.682, $p < 0.001$): The baseline log-odds of admission for a Female in Dept A is 0.682, corresponding to an admission probability of approximately 66%, the highest admission rate across all groups.

GenderMale (-0.100, $p = 0.217$): The Wald test is non-significant. There is no evidence that gender affects admission probability once department is accounted for. This confirms the conditional independence finding from the log-linear analysis.

Department effects: All departments show lower admission odds than Dept A, with effects growing progressively stronger from B through F. Only Dept B is non-significant ($p = 0.693$), meaning it has a similar admission rate to Dept A. Depts C, D, E, and F are all highly significant ($p < 2e\text{-}16$), with odds ratios of 0.283, 0.274, 0.175, and 0.037 respectively. Dept F is the most selective by far, applicants there have only about 4% of the admission odds of a Dept A applicant.

Conclusion: The Wald tests paint a clear picture, admission outcomes are almost entirely determined by department, not gender. The apparent gender gap observed marginally disappears once department is controlled suggesting a Simpson's paradox.

c)

```r
# Fitted from binomial
p_hat  <- fitted(model.b)
total  <- new_data$Admitted + new_data$Rejected
fitted_admitted_binom <- p_hat * total
fitted_rejected_binom <- (1 - p_hat) * total

# Fitted from log-linear
loglin_fitted <- data.frame(
  Gender   = data$Gender,
  Dept     = data$Dept,
  Admit    = data$Admit,
  Fitted   = fitted(model.a.2)
)

admitted_loglin <- loglin_fitted[loglin_fitted$Admit == "Admitted", ]
rejected_loglin <- loglin_fitted[loglin_fitted$Admit == "Rejected", ]

# Merge to align with new_data order
admitted_loglin <- merge(new_data[, c("Gender","Dept")], admitted_loglin,
                         by = c("Gender","Dept"))
rejected_loglin <- merge(new_data[, c("Gender","Dept")], rejected_loglin,
                         by = c("Gender","Dept"))

# Final comparison
result <- data.frame(
  Gender          = new_data$Gender,
  Dept            = new_data$Dept,
  binom_admitted  = round(fitted_admitted_binom, 4),
  loglin_admitted = round(admitted_loglin$Fitted, 4),
  binom_rejected  = round(fitted_rejected_binom, 4),
  loglin_rejected = round(rejected_loglin$Fitted, 4)
)

result
```

```
##    Gender Dept binom_admitted loglin_admitted binom_rejected loglin_rejected
## 1  Female    A        71.7301         71.7301        36.2699         36.2699
## 2  Female    B        16.3605         16.3605         8.6395          8.6395
## 3  Female    C       212.7547        212.7547       380.2453        380.2453
## 4  Female    D       131.7926        131.7926       243.2074        243.2074
```

```
## 5  Female    E       101.3192        101.3192       291.6808        291.6808
## 6  Female    F        23.0429         23.0429       317.9571        317.9571
## 7    Male    A       529.2699        529.2699       295.7301        295.7301
## 8    Male    B       353.6395        353.6395       206.3605        206.3605
## 9    Male    C       109.2453        109.2453       215.7547        215.7547
## 10   Male    D       137.2074        137.2074       279.7926        279.7926
## 11   Male    E        45.6808         45.6808       145.3192        145.3192
## 12   Male    F        22.9571         22.9571       350.0429        350.0429
```

We exactly have the same fitted counts for the binomial (b) compared to log-linear (poisson).

d)

```r
# Fitted probabilities
new_data$p_hat <- fitted(model.b)

# Compute odds for each row
new_data$odds <- new_data$p_hat / (1 - new_data$p_hat)

# Separate female and male
odds_female <- new_data[new_data$Gender == "Female", c("Dept", "odds")]
odds_male   <- new_data[new_data$Gender == "Male",   c("Dept", "odds")]

# Merge and compute ratio Female/Male
odds_ratio <- merge(odds_female, odds_male, by = "Dept")
names(odds_ratio)[2:3] <- c("odds_female", "odds_male")
odds_ratio$ratio_F_over_M <- odds_ratio$odds_female / odds_ratio$odds_male

odds_ratio
```

```
##   Dept odds_female  odds_male ratio_F_over_M
## 1    A  1.97767415 1.78970606       1.105027
## 2    B  1.89368289 1.71369775       1.105027
## 3    C  0.55951970 0.50634014       1.105027
## 4    D  0.54189394 0.49038961       1.105027
## 5    E  0.34736324 0.31434809       1.105027
## 6    F  0.07247174 0.06558366       1.105027
```

A réfléchi pendant 1 s They odd ratio (F/M) are equal to 1.105027in every department.

e)

```r
# Get fitted values from conditional independence model (model.a.1)
loglin_fitted <- data.frame(
  Gender = data$Gender,
  Dept   = data$Dept,
  Admit  = data$Admit,
  Fitted = fitted(model.a.1)
```

```
)

# Separate admitted and rejected
admitted <- loglin_fitted[loglin_fitted$Admit == "Admitted", ]
rejected  <- loglin_fitted[loglin_fitted$Admit == "Rejected", ]

# Compute odds = admitted/rejected for each Gender x Dept
merged <- merge(admitted, rejected, by = c("Gender", "Dept"))
merged$odds <- merged$Fitted.x / merged$Fitted.y

# Separate female and male
odds_female <- merged[merged$Gender == "Female", c("Dept", "odds")]
odds_male   <- merged[merged$Gender == "Male",   c("Dept", "odds")]

# Merge and compute ratio Female/Male
odds_ratio <- merge(odds_female, odds_male, by = "Dept")
names(odds_ratio)[2:3] <- c("odds_female", "odds_male")
odds_ratio$ratio_F_over_M <- odds_ratio$odds_female / odds_ratio$odds_male

print(odds_ratio)
```

```
##   Dept odds_female  odds_male ratio_F_over_M
## 1    A  1.81024096 1.81024096             1
## 2    B  1.72093023 1.72093023             1
## 3    C  0.54026846 0.54026846             1
## 4    D  0.51434034 0.51434034             1
## 5    E  0.33638444 0.33638444             1
## 6    F  0.06886228 0.06886228             1
```

The conditional independence model (model.a.1) gives odds ratios of exactly 1 in every department, gender and admission are completely unrelated within departments by construction, since there is no Gender:Admit term.

The binomial model (model.b) gives a constant ratio of 1.105 across departments, slightly favouring females.

Both are constant across departments, confirming the main effects structure in both cases.